

CSE6344 Theory of Computation

Assignment 1

Topic: NFA, DFA, Regular expressions

Overview

You are to work in groups of 3 or 4 members to develop a DFA that accepts a given set of strings for an unspecified language.

Activities

1. Divide the given set of strings and assign to each team member.
2. Each team member performs the following tasks using JFLAP –
 - Write the regular expression representing the given set
 - Convert the regular expression to NFA
 - Convert the NFA to DFA
 - Test the DFA with accept and reject strings
3. Combine all the DFAs into a single DFA that accepts the given set of strings.
4. Test the DFA with accept and reject strings
5. Document all the work in a report.

Deliverables

- Report containing:
 - How the strings are divided to each member
 - Documentation of individual member's work (regular expressions, NFAs, DFAs)
 - Documentation of the final DFA
- JFLAP file for each member
- JFLAP file for the final DFA

Due Date:

Friday 19 December 2025 (Report)

Friday 26 December 2025 (Demonstration)

Assessment (20 marks)

The assignment will be assessed based on the following criteria:

- Group report – description of assignment work and combination of DFAs
- Group presentation
 - a. Online presentation or video recording
 - b. Explanation of regular expressions, NFA, DFA, and demonstration by every group member

COGNITIVE COMPONENT RUBRIC PROJECT REPORT

Descriptive Elements	Very Weak (1)	Weak(2)	Fair(3)	Good(4)	Excellent(5)
Regular expressions, NFA, DFA	Items are poorly written and is difficult to understand	Items are written with limited clarify and coherence and require further improvements	Items are written fairly coherently and clearly but require minor improvements	Items are written coherently and clearly	Items are well written with excellent coherence and clarity
Combined DFA	Items are poorly written and is difficult to understand	Items are written with limited clarify and coherence and require further improvements	Items are written fairly coherently and clearly but require minor improvements	Items are written coherently and clearly	Items are well written with excellent coherence and clarity
Report contents and quality	Report is poorly written and is difficult to understand	Report is written with limited clarify and coherence and require further improvements	Report is written fairly coherently and clearly but require minor improvements	Report is written coherently and clearly	Report is well written with excellent coherence and clarity

Note for Rate: 0 = no submission

AFFECTIVE COMPONENT RUBRIC PRESENTATION

Descriptive Elements	Very Weak (1)	Weak(2)	Fair(3)	Good(4)	Excellent(5)
Demonstration & Explanation	Not able to demonstrate and explain clearly and require major improvements	Able to demonstrate and explain but require further improvements	Able to demonstrate and explain fairly clearly and require minor improvements	Able to demonstrate and explain clearly	Able to demonstrate and explain with great clarity

Note for Rate: 0 = no submission